

EVSP–DR: where the research stands

Updated from the 16 September 2026, 23:44 EDT collection. The 128/102/67/35 audit counts remain frozen; new seed results are reported separately.

New proof: [Time-only fleet bounds for all 102 instances](#) · [Full-Partille job and memory rationale](#).

In one paragraph

We optimize electric-bus schedules and charging for Transdev's Partille network (GIRO data) with column generation (CG) and a MIP, under time-varying electricity prices. **Fleet size:** in 93 of 102 test instances, our lower bound equals GIRO's own bus count, so GIRO's fleet is optimal under the continuous baseline there; we have matched that bound with a valid schedule in 67 of 102 while 35 event-grid fleet gaps remain open. In 9 instances the bound is one bus *below* GIRO — the only place our method could beat GIRO on fleet — and none has been closed yet. **Charging cost:** on one 5-duty instance, re-optimizing charging alone cuts GIRO's electricity bill by ~44%; also re-optimizing which bus does which trip saves a further 2–7%. Both numbers were obtained without GIRO's 15% reserve and 3-minute charging rules, and a rerun with those rules is in progress. **Next 48 hours** decide three things: whether the 1-bus savings are real, whether the charging gain survives realistic rules, and whether the warm-start method matters or the MIP just needed more time.

Question 1 — Can we recover GIRO's fleet, and can we beat it?

Latest sequential reach: all six chains now have a 32-bus result at target 32, including selected longer searches and seed repeats. Original one-hour MIPs reached largest targets 26/28/31/29/26/28. These are largest observed matches, not success at every smaller k . New seed results pass individual-route replay; their separate dispatch-conversion audit is pending. The 128/102/67/35 table below is the frozen earlier audit, not a refreshed total. Mixed-model one-bus savings remain unproven.

	Count
Instances (6 chains $\times$ $k = 16 \dots 32$)	102
Lower bound = GIRO's k	93
Lower bound = $k - 1$ (model could beat GIRO)	9
Bound matched by a validated schedule (including 3 h searches)	67
GIRO fleet count matched (a different test)	70
Open gap to the lower bound	35

Open: bound k , best schedule $k+1$ or worse	26
Open: bound $k-1$, best schedule k or worse	9

Every one of the 128 schedules passes physical replay. Duplicate trip copies are driven empty; this leaves fleet and cost unchanged. The bounds are numerical certificates on the discretized model (2.5 kWh / 5 min), not exact-arithmetic proofs.

Important context: all 42 GIRO duties are feasible in our model once their charging is re-planned (0 of 42 are feasible with GIRO's own charging plan, under the baseline 240 kW replay). This establishes a continuous-charging k -bus upper bound. It does not itself establish representability on our event grid; that requires a separate feasible event-grid schedule.

Vehicle-group result: time-only scheduling with deadheading needs exactly GIRO's fleet in all 102 instances when groups stay separate, for each group individually. With mixing allowed, the time-only minimum is k in 93 cases and $k-1$ in the same nine as our LP results. Thus group separation rules out those one-bus savings, independently of energy constraints. The electric-bus integer saving with mixing is still open. [All102 saved LP endpoints](#).

Open: can an integer electric-bus schedule achieve the one-bus saving when vehicle groups may mix? The six single-factor arms also test the effects of depot power, reserve and battery size.

Question 2 — Does warm-starting from the previous instance matter?

Updated 16 September, 23:44 EDT: sequential construction remains much stronger in the tested comparisons. All 18 fresh- $k15$ repeats missed the target after 3 hours of fleet search; their integer gaps remain open.

In the original accumulated-CG-time comparison, warm runs matched 24/24 targets and fresh runs 6/24: 5/6 at $k=5$, 1/6 at $k=8$, none at $k=10$ or 15. Four fresh pools provably exclude their target; other gaps remain open. All 24 fresh CGs reached their pricing stopping certificate. The completed $k15$ repeats use 16–19 buses, with a pool bound near 15, across three seeds per chain. This is evidence for the tested sequential method, not proof that fresh pricing can never succeed. Also running: a chain built from *random* trip groups rather than GIRO's duties, to test whether warm-starting helps without knowing GIRO's grouping.

Question 3 — What is demand response worth?

On one 5-duty instance, charging reoptimization produces most of the savings; fresh-CG-derived duties add 2–7% under relaxed rules. A preliminary morning comparison retains a 1.93% gain with reserve and minimum-charge rules. Full GIRO constraints and larger-size gains remain unestablished.

Tariff peak	GIRO as-is	Same duties, optimized charging	Optimized duties + charging
08:00	230.3–231.0	128.3	124.7

12:00	289.6–290.6	164.2	160.6
18:00	223.4–223.7	95.3	88.3

Five buses in every arm; ending energy matched between the two optimized arms. **Caveats that matter:** both optimized arms run batteries to ~0% (GIRO never goes below 19%) and use charges as short as 26 seconds (GIRO's minimum is 3 min). Part of the 44% is spending GIRO's safety margin. Settings: 240 kWh / 350 kW / no start fee / flat depot rate.

Running: Stricter k5 check (15% reserve, 3-minute minimum, still 350 kW): morning costs are 157.965 fixed-duty versus 154.921 fresh CG, 1.93% lower. Both serve trips exactly once; fresh CG returns with more total energy. Evening shows a 2.89% lower cost but retains one duplicate trip pending its dispatch check. The noon fixed-duty job timed out before its first worker record. Larger k15 comparisons remain under review.

Question 4 — Which algorithm changes helped?

Event-based time/SOC graphs with exact pricing (summer) give certified LP bounds where the old capped search could not. Indexed route replay: –12 to –18% CG time. Skipping unused LP-matrix construction: –9 to –15%. Inheriting the full column pool instead of 512 routes: –41 to –65% CG time and better fleets (9→8, 11→10, 17→15) at the same LP value. Three inputs, two orders each — directional, not statistical. Capacity-pricing caching: no measurable effect (both arms time out).

Our model vs GIRO's reality

	Our baseline	GIRO / Transdev	Status
Battery	240 kWh, one type	236.4 / 239.0 kWh, two vehicle groups	Negligible
Terminal charging	240 kW constant	371→120 kW, SOC-dependent	Ours is slower on average
Depot charging	240 kW	60 kW	Ours is 4× faster — testing (item 11)
Minimum SOC	none	15% hard	Testing (k5 rerun, item 11)
Min charge duration	none	3 min + 45 s setup	Testing (k5 rerun, item 10)
Vehicle groups	mixed	route-21 blocks separate	Testing (item 11)
Shared charger capacity	unlimited	1–2 per station, FIFO at 4808	Not modeled
Tariff in scaling runs	flat	—	k=15 tariff runs in progress

Running now

Times are EDT. For running jobs, Expected gives the Slurm allocation deadline (start + wall limit). Sequential projections use the remaining dependency chain at full wall limits, with zero queue delay; they are not scheduled finish times.

Experiment	Answers	Jobs	Expected
Fresh k=15 pools, 3 h fleet search × 3 seeds	Q2: is the 6/24 an artifact of MIP time?	18	Complete: 0/18 target matches; 16–19 buses found.
Chain 5 k=31, 12 h search on existing pool	Q1: can we find 30 buses where GIRO has 31?	1	17 Sep, 07:22
k=32 repeats × 3 seeds, chains 1/3/4/5	Q1: how much is seed luck?	12	Complete: 12 seed results; new C1/C3 seeds reach 32.
k=5 three-arm with 15% reserve + 3-min	Q3: does the DR gain survive real rules?	6	5/6 results; noon fixed-duty startup timed out.
k=15 × (3 synthetic + 1 real tariff), CG vs fixed-duty	Q3: does the gain hold at scale?	48	17 Sep, 03:04 fixed / 07:04 fresh CG
Chain 5 strict physics (60 kW depot, 15%, groups) to k=31	Q1: are the 9 cases an artifact?	62 sequential	21 Sep, 13:24 — full-budget dependency projection
Random-trip-group chain to 364 trips	Q2: does warm start need GIRO's grouping?	42 sequential	17 Sep, 22:37 — full-budget dependency projection
Full Partille, 40 duties / 948 trips, 12 h allocation	Scale	CG 343119: Scaglione, 120 GiB; 11 h 45 min CG; checkpoint fallback tested	Waiting for graph 341404_0; no scheduled start
Frölunda ladder	Generalization	1 (48 h)	18 Sep, 18:12
C5 k31 single-factor arms, full-pool initialization	Q1: isolate depot power, reserve, battery size and vehicle groups	6 arms / 750 replay tasks	Replay started 16 Sep, 20:22; full-pipeline finish not scheduled

What is still needed before writing up

Must have

1. Q1: group separation now rules out $k-1$ in all nine cases. Still open: a validated electric-bus schedule attaining $k-1$ when groups may mix, and sensitivity to the tighter physics. (Items 8, 11.)
2. Q3 under GIRO's rules: the $k=5$ three-arm with 15% reserve and 3-min minimum. If the gain collapses, the DR story changes. (Running.)
3. Q3 at scale: at least $k=15$, at *equal fleet*, on ≥ 3 chains. Needs the $k=15$ tariff CGs plus a follow-up MIP with fleet cap = k . (Running + planned.)
4. Q2: the 3-hour fresh- $k15$ searches are complete (0/18 matches). Next distinguish missing useful integer routes from unfinished MIP search; all 18 fleet gaps remain open.
5. One large instance: full Partille certified or bounded. (CG queued after graph preparation.)

Should have

6. Shared charger capacity at ≥ 1 station in the $k=15$ runs — the one GIRO constraint we have never modeled.
7. Frölunda beyond $k=2$.
8. A real price series in the headline (currently internal-only under license; need a publishable alternative such as Nord Pool day-ahead if terms allow).

Decisions pending

- Cluster: six full single-factor arms are now launched. Other new submissions remain on hold until the fresh- $k15$, C5 $k31$, constrained- $k5$ and $k32$ seed results are complete.
- Full-Partille CG 343119 is queued on Scaglione with 120 GiB and a 12 h allocation (11 h 45 min CG plus shutdown allowance), after the existing graph job. Memory sizing uses the fresh single-process run, not the older eight-worker inheritance peaks. The downstream MIP remains held.

Sources

Chain table with bounds:

[outputs/independent_review_20260916/execution/audited_chain_results.cs](#)

v. Verdicts and evidence: [execution/README.md](#). Independent review: [REVIEW.md](#).

Charging three-arm: [execution/f6/](#). GIRO replay: [execution/f4/](#). All hashes and job IDs in [execution/ledger.json](#).

[F1–F9 verdicts and execution ledger](#) · [Chain results with numerical lower bounds](#) · [Independent review](#)

[Figures with explanations](#) · [CG curves and bus schedules](#) · [Historical research log](#)